EXPERIMENT NO. : 9 (MAIN)

DATE: 21/10/2026

1. AIM

To design and simulate a 6T SRAM cell using 180nm CMOS technology, determine the Cell Ratio (CR) and Pull-up Ratio (PR), and analyze the read and write operations through transient simulation.

2. TOOL

Cadence Virtuoso

3. THEORY

1. 6T SRAM Cell Architecture: A standard six-transistor static random access memory (6T SRAM) cell consists of two cross-coupled CMOS inverters forming a bistable latch that retains a single binary bit at complementary nodes (Q and $\overline{Q}$). Two NMOS access transistors connect the internal storage nodes to complementary bitlines (BL and $\overline{BL}$) under the control of a common Wordline (WL). The cross-coupled topology provides static data retention through positive regenerative feedback as long as the power supply is maintained.

2. Read Operation and Cell Ratio (CR): Prior to read access, complementary bitlines are precharged to V_{DD}. When the Wordline is asserted high, both access transistors turn on. For a cell storing logic '0' ($Q = 0\text{V}$), charge flows from the precharged bitline BL through the access transistor and the driver (pull-down) NMOS to ground, developing a sensing differential ΔV_{BL}. This creates a resistive voltage divider between the access transistor and the driver transistor. To guarantee read stability and avoid destructive read upsets, the voltage bump at node Q must remain strictly below the threshold voltage of the opposing driver NMOS ($V_{th,n}$). This requirement dictates that the driver transistor must be significantly stronger than the access transistor, defined by the Cell Ratio:
$CR = \frac{(W/L)_{\text{driver}}}{(W/L)_{\text{access}}} = \frac{W_n/L_n}{W_a/L_a} \geq 1.2.$

3. Write Operation and Pull-up Ratio (PR): During a write operation, the bitlines are driven by external write buffers to complementary logic states representing the input data, and WL is pulsed high. To successfully write a logic '0' into a node holding '1', the access transistor must overpower the pull-up PMOS transistor and pull the internal node voltage below the inverter switching threshold, triggering regenerative latch switching. To guarantee writeability, the pull-up PMOS must be weaker than the access NMOS, defined by the Pull-up Ratio: $PR = \frac{(W/L)_{\text{pullup}}}{(W/L)_{\text{access}}} = \frac{W_p/L_p}{W_a/L_a} \leq 0.6.$

4. Data Retention (Hold Mode): In standby mode, the Wordline is held at logic low ($WL = 0\text{V}$), turning off the access transistors and isolating the storage latch from the bitline capacitances. The positive feedback continuously maintains the latched complementary voltages with only minute static leakage currents.

4. TABLES

Table 4.4: Design & Simulation Parameters

Parameter	Value	Unit
Technology	180	nm
Supply Voltage (V_{DD})	1.8	V
Channel Length (L)	180	nm
Access Transistor Width (W_a)	800	nm
Driver NMOS Width (W_n)	9.6	μm
Pull-up PMOS Width (W_p)	400	nm
Bitline Capacitance (C_{BL})	100	fF
Simulation Stop Time	40	ns

Table 4.4: SRAM Sizing Ratios & Stability Criteria

Metric	Formula	Value	Stability Criterion	Status
Cell Ratio (CR)	W_n/W_a	12.00	$CR \geq 1.2$	Satisfied (High Read Stability)
Pull-up Ratio (PR)	W_p/W_a	0.50	$PR \leq 0.6$	Satisfied (Reliable Writeability)

Table 4.4: SRAM Operation & Transient Timing Analysis

Time Interval	Operation Mode	WL	BL	$\overline{BL}$	SWS	Node Q	Node $\overline{Q}$	Observation
0.0 to 2.7 ns	Retention (Hold)	1.8 V	0.0 V	1.8 V	1.8 V	0.0 V	1.8 V	Initial state '0' maintained stably
2.7 to 6.7 ns	Write '1' Operation	1.8 V	1.8 V	0.0 V	1.8 V	0.0 V to 1.8 V	1.8 V to 0.0 V	Q flips to '1' at t = 4.0 ns
6.7 to 8.0 ns	Precharge & Hold	0.0 V	Precharge	Precharge	1.8 V	1.8 V	0.0 V	Internal data '1' held securely
8.0 to 11.5 ns	Write '0' Operation	1.8 V	0.0 V	1.8 V	1.8 V	1.8 V to 0.0 V	0.0 V to 1.8 V	Q flips back to '0' at t = 8.0 ns
11.5 to 16.0 ns	Read '0' Operation	1.8 V	Discharging	1.8 V	0.0 V	0.0 V	1.8 V	BL discharges; no read upset occurs

5. CALCULATION

$$CR = W_n/W_a = 9.6/0.8 = 12$$

$$PR = W_p/W_a = 0.4/0.8 = 0.5$$

6. CIRCUIT DIAGRAMS & WAVEFORMS

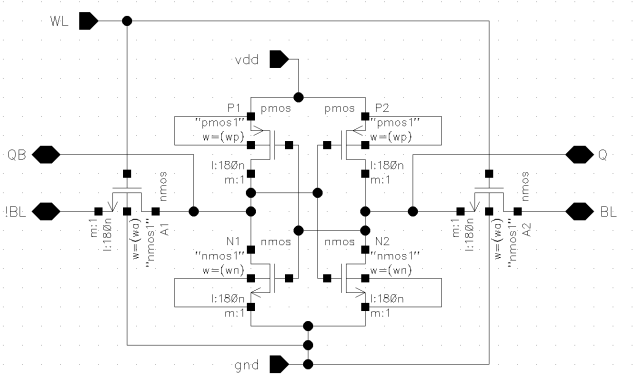


Figure 6.1: 6T SRAM cell schematic ($W_n = 9.6\,\mu m, W_p = 0.4\,\mu m, W_a = 0.8\,\mu m$)

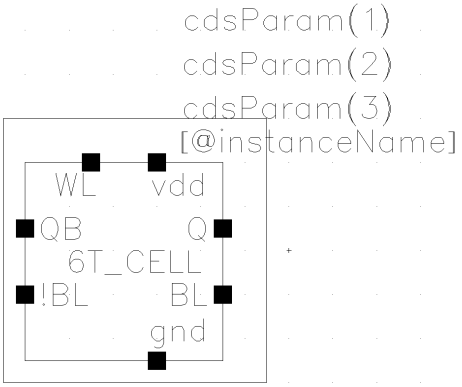


Figure 6.2: 6T SRAM cell symbol ($WL, BL, \overline{BL}, Q, \overline{Q}$)

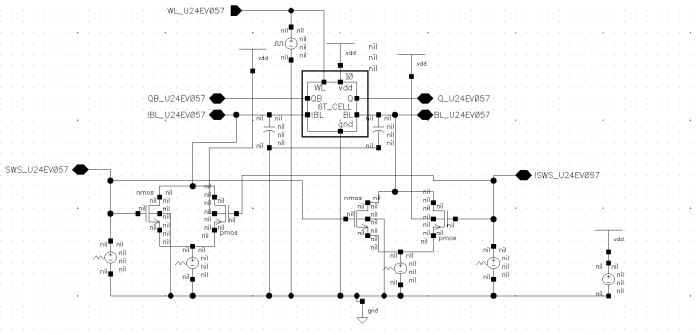


Figure 6.3: 6T SRAM cell testbench schematic with write drivers and bitline loads

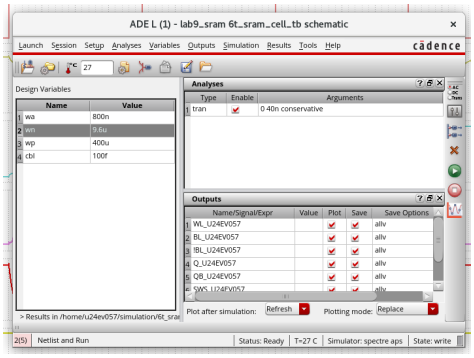


Figure 6.4: ADE L simulation state & design variable setup

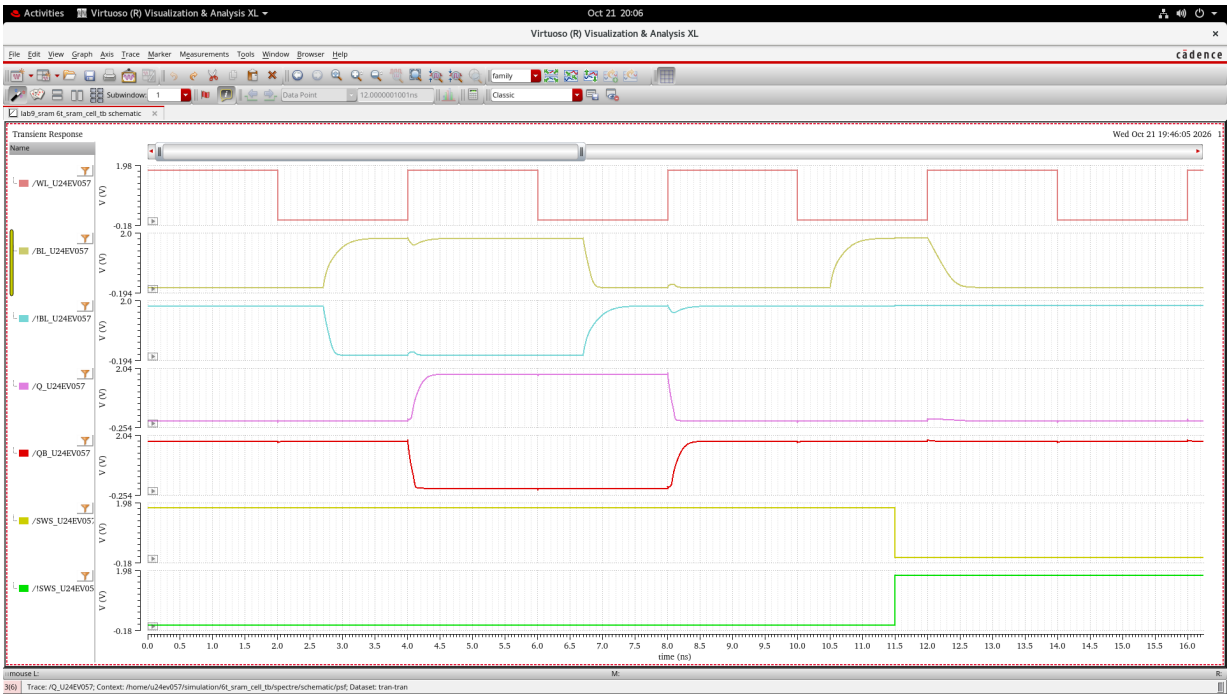


Figure 6.5: Transient waveforms demonstrating Write '1', Retention, Write '0', and Read '0' operations

Signals: WL (Wordline), $BL, \overline{BL}$ (Bitlines), $Q, \overline{Q}$ (Stored state), $SWS, \overline{SWS}$ (Write driver control)

7. CONCLUSION

1. The 6T SRAM cell was successfully designed and simulated in 180nm CMOS technology, achieving a Cell Ratio ($CR = 12.00$) and Pull-up Ratio ($PR = 0.50$) that satisfy the read-stability and writeability criteria.

2. Transient simulations verified robust functionality across all states: successfully demonstrating Write '1', data retention, Write '0', and non-destructive differential Read '0' without read disturbance.